



The Norwegian Origin Asset

Acquisition, Analysis, and Strategic Po-
sition of US Patent 11,803,853 B2

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Lux Industries Inc.

Internal strategic memorandum

Abstract

In late 2025, Lux Industries Inc. acquired US Patent 11,803,853 B2 (“Controlling transactions on a network”) from Nahmii AS, a Norwegian state-channel and Layer-2 scaling company. The patent, granted October 2023 with a priority date of 4 May 2018, claims an architecture in which a server originates blockchain transaction records, an independent set of validating nodes determines completeness and correctness of those records, and a smart-contract execution agent on a parent blockchain permits or denies execution based on the validation result. The claim language reads broadly on virtually every modern Layer-2 and cross-chain protocol: optimistic and validity rollups, Alt-DA architectures, cross-chain messaging systems (LayerZero, Chainlink CCIP, Axelar), and Rollups-as-a-Service platforms. This paper documents the patent’s content, the architectural insights underlying its breadth, the commercial significance of its claim scope, the prior-art landscape and resulting invalidity risk, and the strategic position of Lux Industries as patent owner. The paper concludes that the patent, properly prosecuted and enforced, supports a 5-7 year monetization program with expected realized value of \$500M-\$2B across the OP Stack, Arbitrum Orbit, Polygon CDK, and cross-chain messaging ecosystems, while serving as the foundational asset of a defensive substrate-licensing program that positions Lux as the licensed-rollup standards body for institutional and regulated tokenization.

1 Provenance of the patent

1.1 Nahmii AS and the hubii network

Nahmii AS was a Norwegian blockchain company founded as “hubii network” in 2017 by a team based in Stavanger, Norway. The company built a payment-channel-derived Layer-2 scaling system on Ethereum and tokenized its native asset under the ticker HBT. The technical work was substantial: the company shipped a production Layer-2 system with on-chain dispute resolution, a challenge bot continuously monitoring for fraud, and a complete settlement protocol named “driip” (a portmanteau of “digital” and “trip”). The code remains publicly accessible at github.com/hubiinetwork, including the contract suite (`nahmii-contracts`), the challenge bot (`nahmii-challenge-bot`), and the user-side tooling (`nahmii-sdk`, `hubii-core`).

Nahmii AS ceased active development around 2020. The repositories show last meaningful technical commits in February 2020 (`nahmii-contracts` v1.5.0 release; `nahmii-challenge-bot` final feature merge), with subsequent activity limited to a 2021 SDK peer-dependency update and 2024 exit-documentation updates instructing users how to withdraw remaining funds. The company wound

down operations while the patent application, filed 3 May 2019, continued through prosecution.

1.2 The patent issues post-shutdown

The patent was granted 31 October 2023, three years after Nahmii ceased active development. This is a structurally significant fact. A patent granted to a wound-down entity, in an actively-infringing space, with no ongoing enforcement, is the canonical monetization-ready asset — the entity holds the patent without an operating business that would deprioritize enforcement; the technology space the patent reads on has scaled to billions in commercial activity; and the absence of public enforcement signals that the asset was awaiting either a strategic acquirer or sale to a patent monetization vehicle.

1.3 Acquisition by Lux Industries Inc.

Lux Industries Inc., a Delaware corporation, acquired the patent in late 2025 through a transaction executed with the residual entity controlling Nahmii AS’s intellectual property. The acquisition includes:

- Full ownership of US Patent 11,803,853 B2, recorded with the USPTO.
- All foreign counterparts (subject to ongoing diligence to inventory).
- Rights to file continuations, divisionals, and reissues on the parent.
- Inventor consents required for any prosecution actions.
- Verification of clean chain of title back through Nahmii AS’s Norwegian dissolution process to ensure no creditor encumbrances.

The acquisition is non-public. Lux Industries holds the asset within its general intellectual property holdings and prosecutes new applications under its own name claiming priority where applicable.

2 What the patent actually claims

2.1 The architectural premise

The patent describes a network architecture for controlling blockchain transaction execution in which the central problem is operator trust. The patent’s premise: how does a blockchain system achieve high throughput by delegating transaction origination to an off-chain operator, while preventing that operator from cheating?

The answer the patent advances: *partition transaction origination from transaction execution, and insert an independent validation layer between them*. The operator originates the transaction record; an independent set of validating nodes checks the record for completeness and correctness; an on-chain smart contract executes the transaction only after the validation produces an acceptance result.

This is the conceptual foundation of every modern Layer-2 rollup, every Alt-DA architecture, every cross-chain bridge with a verifier network, and every regulated-tokenization stack with a transfer-agent or compliance gate. The architectural insight is broad enough to capture most of the structural variants of the blockchain scaling landscape.

2.2 The independent claim elements

The patent contains three independent claims (claim 1 is the system claim, claim 10 is the method claim, claim 18 is the computer-readable medium claim). The element structure across all three is co-extensive. The element list for claim 1:

El	Element (paraphrased)	Construction notes
1a	A first set of nodes configured to generate transactions between a first and a second participant	Loose: any pair of transacting parties. Construction is patentee-favorable.
1b	A server configured to originate a transactional record for the transaction	“Originate” is the load-bearing word. The construction battle.
1c	A second set of nodes configured to validate the transactional record	The validator/verifier network. Plasma’s watchers; CCIP’s Commit DON; LayerZero’s DVNs; Axelar’s validators; OP Stack’s challengers.
1d	Validation includes checking completeness (data availability) of the record	Patent expressly distinguishes completeness from correctness.
1e	Validation includes checking correctness (fraud / validity) of the record	Same.
1f	A first agent (smart contract on parent chain) configured to receive a request to execute the transaction	The destination-chain execution contract.
1g	A second agent configured to obtain validation results from the second set of nodes and provide them to the first agent	The oracle / verification helper.
1h	First agent permits or denies execution based on validation result	The execution gate.
1i	A threshold number of nodes in the second set substantiates completeness or correctness	The quorum mechanic.

A determined defendant will fight construction on elements 1b, 1c, and 1g. The patentee’s task is to obtain claim construction at Markman that (i) reads element 1b broadly enough to capture sequencers, batchers, coordinators, and OP-Stack-style proposers; (ii) reads element 1c broadly enough to capture DVN/DON/validator-network/DA-committee architectures; (iii) reads element 1g broadly enough to recognize the two-contract pattern (commit-store-plus-RMN; ULN-plus-EndpointV2; AxelarAuthWeighted-plus-Gateway) as the first-agent/second-agent pair.

2.3 What the specification says

The specification’s named embodiment maps the claim elements as follows: server $\equiv$ state-channel operator (Nahmii’s “driip operator”); second set of nodes $\equiv$ peer-to-peer validators (Nahmii’s challenge-bot operators); first agent $\equiv$ Ethereum smart contract; second agent $\equiv$ Chainlink-style oracle helper. The specification therefore explicitly contemplates a Chainlink-class oracle as the second agent. This bridge from the on-chain execution contract to the off-chain validation network is the key insight that makes the patent read on modern oracle-mediated bridge architectures.

3 The architectural breadth and why it matters

The patent’s value derives from the breadth with which it reads on modern infrastructure. We enumerate the major surfaces.

3.1 Layer-2 rollups

Optimistic rollups (Optimism, Base, Arbitrum One). The sequencer is the server; the L2OutputOracle or RollupCore is the second agent maintaining state-commitment publications; the fault-proof challenger network is the second set of nodes; the OptimismPortal or Outbox is the first agent; the dispute game outcome gates withdrawals. Reads, post-Cannon (June 2024), on every element. The Ethereum-DA defense applies only to element 1d (completeness/DA), and is partial: even Ethereum-DA rollups have challengers verifying canonical state derivation. Moderate-to-strong read.

Alt-DA rollups (Mantle, Plasma chains, OP-Stack-with-EigenDA, OP-Stack-with-Celestia, OP-Stack-with-Avail, OP-Stack-with-S3). The DA committee is the second set of nodes producing availability attestations. Element 1d reads cleanly — the committee explicitly attests to availability, and the L1 bridge contract gates withdrawals on the attestation. Strong read across every element. This is the strongest rollup category for enforcement.

Validity rollups / ZK rollups (zkSync, StarkNet, Linea, Scroll, Polygon zkEVM). A validity proof replaces the validator network. Element 1c is weaker — a math proof verified by an L1 verifier is arguably not a “second set of nodes substantiating correctness.” Distinguishable on narrow construction; reads under BRI. Lower-priority but worth charting.

Sovereign rollups and Layer-3 chains. Same architectural pattern as Alt-DA rollups. Strong read.

3.2 Cross-chain messaging

Chainlink CCIP. Commit DON + RMN. Cleanest element-by-element fit among major protocols. RMN explicitly distinguishes data-availability monitoring (element 1d) from correctness/fraud monitoring (element 1e). OffRamp gates execution on CommitStore verification plus RMN non-curse status. Strong read across every element.

LayerZero V2. DVN verification + Executor delivery. SendUln302 originates the Packet (element 1b); DVNs are the verifier network (element 1c); ReceiveUln302 is the second agent (element 1g); EndpointV2 is the first agent gating delivery (element 1f, 1h). The OApp configures required and optional DVN thresholds as explicit protocol parameters (element 1i). Strong read.

Axelar GMP. Validator quorum signature gates command execution at the Gateway. Element 1b (server) is the construction battle since validators perform both origination and validation; argue function-level separation, or pivot to Squid/ITS where the coordinator is distinct. Medium-strong read.

Wormhole. 19-of-19 guardian network attests to source-chain events; destination CoreBridge gates message verification. Similar structure to Axelar. Strong read on the original guardian model; weaker on Wormhole NTT / native token transfer variants.

Hyperlane. ISM (Interchain Security Module) abstraction explicitly separates verification from execution. The ISM is the second agent gating destination-Mailbox delivery on verifier signatures.

Strong read.

3.3 Rollups-as-a-Service platforms

Alchemy Rollups, AltLayer, Caldera, Conduit, Gelato Rollups. RaaS providers operate, host, and configure rollups on behalf of customers. Where their customer deployments use AnyTrust / DAC / Alt-DA, the RaaS provider is the direct, inducing, and contributory infringer. Single license at the RaaS-platform tier protects all customer deployments. Strong commercial-leverage target.

3.4 Institutional tokenization

Tokenized money-market funds (BUIDL, BENJI, USDY, OUSG, sBUIDL, USDL), tokenized securities ATs, regulated stablecoins, CBDCs. Where the tokenization stack uses a transfer agent, custody MPC, and compliance gate to authorize on-chain operations, the three-party authorization architecture maps to the patent. The transfer-agent service is the server (1b); the validator set is the three authorities (1c); the compliance contract is the second agent (1g); the token contract is the first agent (1f, 1h). Direct read on the architecture pattern.

4 Prior art and invalidity risk

The patent's commercial value and its claim breadth attract corresponding invalidity exposure. Three classes of prior art matter.

4.1 Plasma and Minimal Viable Plasma

Plasma white paper (Poon, Buterin; 11 August 2017; 47 pages). Published nine months before the patent's 4 May 2018 priority date. Describes a hierarchy of off-chain Plasma chains each run by an operator, periodically committing Merkle state roots to Ethereum, with fraud proofs and mass-exit safety nets. Plasma §6 explicitly discusses fraud proofs (element 1e); Plasma §5.1 expressly identifies data availability as the defining problem of Plasma and discusses the mass-exit response when the operator withholds data (elements 1d and 1h).

Minimal Viable Plasma (Buterin, Floersch; 9 January 2018; ethresear.ch). Published four months pre-priority. Concrete operator chain with 7-day exit window, confirm signatures, priority queue. Cleanly maps to elements 1a-1h.

These two references are the strongest §102 anticipation challenges. In any IPR, the petitioner will cite both. The Lux response must construct claims narrowly enough to defeat anticipation while reading broadly enough to capture modern infringers. The construction angle: (i) threshold formalization (Plasma's 1-of-N honest assumption is qualitative; the patent's element 1i is a numerical threshold parameter); (ii) explicit completeness/correctness dichotomy (Plasma collapses these; the patent expressly partitions them); (iii) two-agent structure (Plasma has one contract role; the patent's first-agent + second-agent partition is explicit and unique).

4.2 Truebit and interactive verification

Truebit (Deutsch, Reitwießner; December 2017). Interactive verification game for off-chain computation. Provides the interactive-challenge variant of fraud proofs and predates Cannon-style

fault-proof bisection by years. Combinable with Plasma + MVP for §103 obviousness on Cannon-class fault-proof embodiments.

4.3 Chainlink and oracle networks

Chainlink white paper (Nazarov, Ellis, Juels; September 2017). Decentralized oracle network providing validation results to L1 smart contracts. Predates priority by eight months. Provides the “second agent” element trivially. Combinable with Plasma + MVP for §103 obviousness on the oracle-mediated variant of the architecture.

4.4 Overall invalidity exposure

Realistic IPR outcome with strong claim construction: 30-40% probability of total claim cancellation; 40-60% probability of narrow claim survival; remainder split between substitute claims and various partial outcomes. The continuation strategy provides backstop: even if the parent claims are invalidated, continuation claims filed on the same priority date but with narrower, post-Plasma-aware language preserve the enforcement asset.

This is materially better than the conventional wisdom of “software patent in IPR = zero.” The patent’s claim language, properly construed, distinguishes over Plasma/MVP on multiple specific elements. The risk is real but manageable.

5 The commercial significance

5.1 Total addressable infringement universe

The patent reads on infrastructure carrying multi-trillion-dollar cumulative value. As of analysis date:

Surface	TVL / cumulative volume (approximate)
Coinbase Base (OP Stack)	\$10-15B TVL
Optimism Mainnet	\$5-8B TVL
Arbitrum One	\$15-20B TVL
Mantle (OP Stack + DA committee)	\$2-3B TVL
zkSync, Linea, Scroll, Polygon zkEVM combined	\$3-5B TVL
Other Arbitrum Orbit + OP Stack chains (200+)	\$5-10B TVL collective
Chainlink CCIP (cumulative volume, 2023-2025)	\$15B+
LayerZero (cumulative messages)	\$50B+ underlying value
Axelar GMP (cumulative volume)	\$10B+
Wormhole (cumulative volume)	\$50B+ underlying value
Hyperlane and other ISMs	\$5B+
Tokenized institutional finance (BUIDL, USDY, sBUIDL, etc.)	\$10-15B AUM
Total infringement footprint	\$200B+ exposed

Even at a conservative 0.25% running royalty applied selectively to infringers willing to license, the patent supports a multi-hundred-million-dollar annual revenue baseline.

5.2 Realistic monetization model

Litigation campaign across 5-7 years, with target sequencing per the enforcement-targets strategy document:

Phase	Targets	
First-wave settlements (Tier 1: CCIP, LayerZero, Arbitrum Orbit AnyTrust, Alchemy)	4	\$80-30
Second-wave settlements (Tier 2: OP Stack + Alt-DA, other RaaS)	5-10	\$50-20
Third-wave settlements (Tier 3: Axelar, institutional tokenization)	5-15	\$100-50
Tier-4 escalation (Coinbase, OP Foundation, Arbitrum Foundation)	3	\$300-80
Total 5-year EV (probability-weighted)		\$500M-\$1
Probability of total loss (full invalidation, no recoveries)		15-5

5.3 The substrate-licensing alternative

A pure assertion campaign is one mode of monetization. The alternative, more valuable mode is substrate licensing: Lux markets Lux-Network as the legally-licensed, regulatory-compliant rollup substrate; ecosystem participants receive the patent license as part of being on Lux; external rollup operators license at Tier 2 or Tier 3. The patent becomes the legal moat that makes Lux structurally analogous to ARM (in mobile chips), Qualcomm (in cellular), or Visa (in payments) for tokenized rollups. The substrate-licensing model converts patent assertion into platform sales backed by an IP moat, and pulls institutional buyers (BlackRock, Fidelity, JPMorgan, Citi) into the Lux platform.

The substrate-licensing model is the higher-value mode but requires Lux to deliver a working substrate (Z-Chain rollup support, Lux-certified rollup program). Pure assertion provides faster cash; substrate-licensing provides larger long-term value. Lux Industries pursues both modes in parallel.

6 Strategic position

Lux Industries' position as owner of this patent provides:

1. **Enforcement asset.** A 14-year monetization window (through 2039 expiry) against a multi-trillion-dollar infringement universe.
2. **Defensive moat.** Ecosystem participants on Lux Network receive free Tier-1 ecosystem licenses, providing IP cover for white-label participants (Liquidity, MLC, VCC, Pars, Osage, Hanzo, Zoo). Future Z-Chain rollup deployments inherit the moat.
3. **Platform leverage.** Substrate-licensing model positions Lux as the institutional-rollup standards body. BlackRock-class licensees pay for competitive moat against tokenization competitors.
4. **Standards-body inheritance.** Lux can shape the licensing landscape for next-generation rollup architecture by selectively granting licenses to favored substrate variants, encouraging convergence on Lux-compatible designs.

The downsides are also clear:

1. **Invalidity risk.** 15-20% probability of complete asset loss in IPR. Mitigated by continuation strategy.
2. **Anti-troll narrative.** Open-source rollup community will frame Lux as a patent troll. Mitigated by substrate-licensing positioning emphasizing technology delivery alongside enforcement.
3. **Defensive-aggregator pressure.** RPX, Unified Patents, COPA, Allied Security Trust will attempt to acquire or invalidate the patent. Mitigated by litigation finance lockup and non-engagement with these entities.

7 Conclusion

The Nahmii patent acquired by Lux Industries is a singular asset in the blockchain IP landscape: a foundational patent on the operator-plus-validator-plus-execution-gate architecture that underlies most modern Layer-2, cross-chain, and tokenization infrastructure, originating from a defunct Norwegian entity that built the canonical embodiment of the patented architecture and then quietly exited the market while the patent prosecuted to grant. Lux Industries' acquisition of the asset positions the company to monetize across a multi-trillion-dollar infringement universe through a combination of pure assertion and substrate licensing, with realistic 5-7 year expected value in the \$500M-\$1.8B range. Properly prosecuted through the continuation strategy and supported by the post-quantum patent family described elsewhere in this paper portfolio, the patent serves both as a revenue-generating enforcement asset and as the foundational legal moat for the Lux Network ecosystem.

Cross-references

- Patent overview: `~/work/lux/patents/acquired/us-11803853-nahmii-rollup-control/patent-overview`
- Enforcement strategy: `~/work/lux/patents/strategy/enforcement-targets.tex`
- Per-target claim charts: `~/work/lux/patents/claim-charts/`
- Nahmii original source: `~/work/hubii/`
- PQ patent family: `~/work/lux/patents/original-filings/`
- PQ future-infringer paper: `../pq-patent-family-future-infringers/`